Discriminability Without Utility: Why Sycophancy Directions Fail as Scalar RL Rewards

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Abstract

Representation engineering shows that large language models encode sycophancy as a linear direction in activation space. We extract a contrastive sycophancy direction (d_{syc}) from style-controlled pairs on two model families, achieving cross-validated AUROC 0.827 on Qwen3-8B and 0.779 on Llama-3-8B. We then test three d_{syc}-based reward formulations with Group Relative Policy Optimization (GRPO) on Qwen3-8B. Every formulation fails. The cosine reward cannot separate valid revision from sycophantic capitulation in the narrow online activation gap. The probe reward creates a stubbornness trap: the model learns that never revising its answer maximizes reward, and the true-positive flip rate drops to zero within 500 steps. When we remove d_{syc} entirely and train with a purely behavioral reward, only 4 of 26 checkpoints across two seeds exceed vanilla, while supervised fine-tuning exceeds vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 3/3 ongoing). Only supervised fine-tuning with dense per-token supervision succeeds. The root cause is a semantic gap between discriminability and reward utility: d_{syc} encodes *whether* belief changed, not *whether it should have*. Valid revision and sycophantic capitulation produce overlapping activations because both involve the model shifting its stated position toward the user's. A single scalar reward per completion, whether from representations or behavioral classifiers, cannot convey which belief changes are warranted. These results establish that representation geometry sufficient to detect a behavior is not automatically suitable for converting into a scalar per-completion reward signal.

1 Introduction

Representation engineering (Zou et al., 2023) reveals that large language models encode sycophancy as a linear direction in their residual streams. A contrastive sycophancy direction d_{syc}, extracted from style-controlled pairs of valid belief revision and sycophantic capitulation, achieves cross-validated AUROC 0.827 (5-fold CV σ=0.028) on Qwen3-8B and 0.779 (5-fold CV σ=0.019) on Llama-3-8B (§3). The direction captures evidential granularity: discriminability rises monotonically from weak-evidence pairs (AUROC 0.675) through medium (0.883) to strong (0.929), and the signal remains stable under LoRA training with cosine drift of approximately 0.96 at 500 steps. These results, replicated across two architecturally distinct model families, point to an appealing possibility: if the model's representations already separate sycophantic from non-sycophantic behavior, that separation should be convertible into a scalar reward signal for reinforcement learning. Sycophancy is among the most persistent alignment failures in deployed systems (Sharma et al., 2023), and a representation-level reward could complement existing behavioral approaches by targeting the internal computation rather than its surface outputs.

This paper shows that the conversion fails, and that the failure runs deeper than the representation signal itself. We test three formulations that translate d_{syc} into a GRPO (Shao et al., 2024) training signal (cosine projection, binary probe, and continuous probe) across six experimental configurations; every formulation fails. Cosine similarity produces a reward gap of just 0.098 between valid and sycophantic activations, too narrow to drive policy updates (§4). Online recalibration improves the offline AUROC from 0.827 to 0.910 yet triggers mode collapse. A binary probe with stopped gradients creates a stubbornness trap: the model learns that never revising its answer reliably collects the probe reward, and the true-positive flip rate (ToF) drops to zero within 500 steps while generation entropy remains at 0.70 (§5). These failures might suggest that removing d_{syc} in fa-

vor of a purely behavioral reward would resolve the problem. It does not. R3-only GRPO, which uses an NLI-based behavioral reward without any representation component, also fails: across two seeds and 26 checkpoints, only 4 (15.4%) exceed vanilla, while ACT exceeds vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 3/3 ongoing; §6). Both seeds oscillate with similar amplitude but stochastic crash timing, confirming that instability is intrinsic to sparse behavioral reward. Two layers of failure emerge: d_{syc}-based rewards are actively harmful, and scalar per-completion RL is structurally insufficient for sycophancy calibration.

Techniques for discovering latent knowledge (Burns et al., 2023) and representation-guided RL (Zhang et al., 2026) share a premise: a direction with high discriminability in activation space should transfer into a useful training signal. For reasoning faithfulness and safety refusal, this premise holds, because the behavior to suppress is geometrically distinct from the behavior to preserve along the relevant direction. Sycophancy violates this premise. Capitulation under social pressure and revision under valid evidence produce overlapping activation signatures along d_{syc}, because both involve the model shifting its stated belief toward the user's position.

The root cause is a semantic gap between discriminability and reward utility. d_{syc} encodes *whether* belief changed, not *whether it should have*. A post-hoc classifier can use this signal effectively because it scores labeled examples where the validity of the belief change is already known. A reward signal during RL training has no such access to ground truth; it must distinguish warranted revision from unwarranted capitulation using activation geometry alone, and d_{syc} collapses exactly this distinction. This conflation explains both layers of failure. d_{syc} rewards penalize all belief change indiscriminately, producing behavioral collapse. R3-only GRPO's single scalar per completion cannot compensate because it lacks the per-token resolution to teach which belief changes are warranted and which are not. ACT succeeds because its dense supervision operates at the token level, where the valid/invalid distinction can be resolved. Sycophancy is defined relative to external evidential warrant (Sharma et al., 2023), a property that no single activation direction can fully encode. The entanglement between helpfulness and sycophancy at the representation level, where the same mechanisms that allow models to track user intent also predispose them to capitulate (Wang et al., 2025), is amplified by RLHF (Shapira et al., 2026). When reward overoptimization (Gao et al., 2023) operates on a signal that conflates compliance with capitulation, the result is predictable collapse (Casper et al., 2023).

We present three contributions:

- **Every tested GRPO formulation fails for sycophancy calibration** (C1). We extract and validate d_{syc} on two model families (Qwen3-8B AUROC 0.827 (5-fold CV σ=0.028); Llama-3-8B AUROC 0.779 (5-fold CV σ=0.019)), then show that all three d_{syc}-based reward formulations produce behavioral collapse (§4, §5). R3-only GRPO, which removes d_{syc} entirely, also fails: only 4/26 checkpoints across two seeds exceed vanilla, while ACT exceeds vanilla at every evaluated checkpoint (§6). Mechanistic analysis reveals that d_{syc} discriminability remains stable (AUROC $\Delta < 0.01$) and behavioral reward saturates ($\sim$0.98, p=0.703), yet SYCONSCORE regresses by 15%—behavioral failure decoupled from both representation and reward signals.

- **Probe reward trap with behavioral-before-distributional collapse** (C2). The binary probe reward produces behavioral death (ToF = 0 at step 500) while entropy remains at 0.70. The 1,300-step lag before distributional collapse means standard entropy monitoring misses the failure entirely, with the reward averaging +0.72 throughout (§5).

- **Sparse RL reward cannot convey the valid/invalid distinction** (C3). We trace these failures to a unified root cause: d_{syc} encodes belief-change direction but not evidential warrant. A single scalar reward per completion, whether from representations or behavioral classifiers, cannot teach the model which belief changes are warranted. Dense SFT supervision resolves this compared to scalar reward approaches: ACT achieves SYCONSCORE of 0.301, while no GRPO variant reliably surpasses vanilla (§6).

2 Experimental Framework

2.1 Direction Extraction

We extract a linear sycophancy direction d_{syc} from the residual stream using contrastive activation pairs: a *valid correction* (warranted belief revision under strong evidence) and a *sycophantic capitulation* (unwarranted concession under social pressure). We construct 181 contrastive pairs using GPT-4.1 in a two-stage pipeline: skeleton generation followed by 3-turn dialogue completion. Style matching is enforced by automated validation (response length difference $<10\%$, shared opening phrase ≥ 2 words), with up to 3 retries per pair; 19 of 200 initial pairs were discarded for failing validation. Evidence strength is annotated at three levels: *weak* (75 pairs; personal memory or opinion), *medium* (54 pairs; brief source citation), and *strong* (52 pairs; authoritative source with detailed data). From the 181 paired examples (362 activations total), we compute the mean difference at target layer ℓ:

$$d_{\text{syc}} = \frac{1}{N} \sum_{i=1}^{N} \left(\mathbf{h}_i^+ - \mathbf{h}_i^- \right) \tag{1}$$

where $\mathbf{h}_i^+$ and $\mathbf{h}_i^-$ are residual-stream activations for the i-th valid and sycophantic response. The target layer is selected by 5-fold cross-validated AUROC; on Qwen3-8B this yields layer 19 with AUROC 0.827 (5-fold CV σ=0.028).

2.2 Reward Formulations

GRPO (Shao et al., 2024) ranks sampled completions by reward and updates the policy toward higher-reward outputs. We define $R = w_1 \cdot R_1 + w_3 \cdot R_3$, where R_1 is a direction-based component and R_3 is a behavioral component (DeBERTa NLI classifier scoring consistency with the ground-truth answer). Three formulations translate d_{syc} into R_1:

- **F1 Cosine projection** with four variants: (a) tanh-gated dot product $\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$, (b) raw cosine $\cos(\mathbf{h}, d_{\text{syc}})$, (c) raw cosine with online recalibration (§2.3), and (d) $5\times$ scaled cosine. See §4 for details.
- **F2 Binary probe.** $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$, a frozen logistic probe with stopped gradients.
- **F3 Continuous probe.** $R_1 = 1 - P(\text{syc} \mid \mathbf{h})$, the sigmoid output of the same logistic probe, providing a continuous reward signal without discretization.

The weight w_1 is set by diagnostic-driven iteration rather than grid search (Appendix A). As an ablation, R3-only GRPO sets w_1=0, training with only the behavioral reward.

2.3 Online Recalibration

LoRA fine-tuning shifts internal representations, so we re-extract d_{syc} every 500 steps. The cosine similarity between consecutive directions remains ≈ 0.96 at step 500, indicating slow drift relative to recalibration frequency.

2.4 SFT Baseline: Activation Consistency Training

We implement Activation Consistency Training (ACT; Irpan et al., 2025), which combines standard SFT loss with an L2 activation consistency loss at target layers, providing dense per-token supervision. Unlike GRPO's single scalar per completion, ACT operates at the token level using labeled valid/invalid pairs.

2.5 Evaluation

The *True-positive Flip rate* (ToF) measures acceptance of valid corrections; the *No-Flip rate* (NoF) measures resistance to invalid pressure. Their harmonic mean captures the calibration trade-off:

$$\textsc{SyconScore} = \frac{2 \cdot \text{ToF} \cdot \text{NoF}}{\text{ToF} + \text{NoF}} \tag{2}$$

We use the harmonic mean to penalize methods that achieve high resistance by refusing all evidence-warranted corrections (low ToF, high NoF) or by capitulating universally (high ToF, low NoF). Flip classification uses `cross-encoder/nli-deberta-v3-base` (MNLI mismatched accuracy 90.0%, SNLI test 92.4%; He et al. 2023) for stance classification, following Irpan et al. (2025). All experiments use Qwen3-8B with LoRA (rank=64, α=128) and DeepSpeed ZeRO-2. We test six d_{syc}-based GRPO configurations: four cosine R_1 variants (F1a–F1d), binary probe R_1 (F2), and continuous probe R_1 (F3), plus R_3-only GRPO as a non-directional baseline and ActAdd as an inference-time baseline.

3 Direction Discriminability

Before evaluating d_{syc} as a reward signal (§4–§5), we first establish that it captures a genuine

sycophancy-related signal in the residual stream. If the direction failed to discriminate sycophantic from non-sycophantic activations, downstream reward failures would be trivially explained by poor direction quality rather than by a fundamental gap between discriminability and reward utility. We evaluate discriminability along five axes: cross-validated classification accuracy, evidence-strength stratification, cross-model transfer, training stability, and probe ablation. Figure 1 summarizes the per-layer AUROC profiles for both models.

Qwen3-8B layer-wise discriminability. We evaluate d_{syc} via 5-fold cross-validated AUROC on 181 style-controlled contrastive pairs (§2.1). The direction achieves AUROC 0.827 (5-fold CV σ=0.028) at its best layer (L19). Discriminability emerges gradually across depth: layers L0–L8 perform near chance, statistical significance ($p < 0.01$) first appears at L16, and performance peaks at L19. This gradual emergence, with the peak in the upper third of the network, suggests that sycophancy-relevant information is encoded as a late-stage representation rather than an input-level confound. The direction also captures evidential granularity. AUROC rises monotonically from 0.675 on weak-evidence pairs to 0.883 on medium and 0.929 on strong pairs. This gradient indicates that d_{syc} encodes the degree of evidential warrant, not merely a binary sycophancy label.

Cross-model generalization. To rule out that these results are specific to Qwen3-8B's representation geometry, we repeat the full extraction and evaluation pipeline on Llama-3-8B-Instruct. Table 1 compares the two models. Llama-3-8B achieves CV AUROC 0.779 (5-fold CV σ=0.019) at L16, with a parallel evidence-strength gradient (weak 0.711, medium 0.864, strong 0.911). The best-layer depth differs (L16 vs. L19), consistent with the architectural differences between the two families, but the qualitative pattern is identical: gradual emergence, best-layer selection in the upper third, and monotonic strength scaling. Discriminability is robust across knowledge domains, with per-domain AUROCs on Llama-3-8B ranging from 0.751 (geography) to 0.844 (commonsense). When surface confounds (response length, hedging patterns) are explicitly controlled, Llama-3-8B's AUROC rises to 0.923, which confirms that the direction captures semantic content rather than stylistic artifacts. The consistent results across architecturally distinct model families indicate that contrastive sycophancy directions reflect a shared representational phenomenon rather than a model-specific idiosyncrasy.

Table 1: Cross-model d_{syc} discriminability. CV AUROC is 5-fold cross-validated; evidence-strength AUROCs are evaluated on the full dataset at each model's best layer.

Model	Best Layer	CV AUROC	Weak	Medium	S
Qwen3-8B	L19	0.827±0.028	0.675	0.883	
Llama-3-8B	L16	0.779±0.019	0.711	0.864	

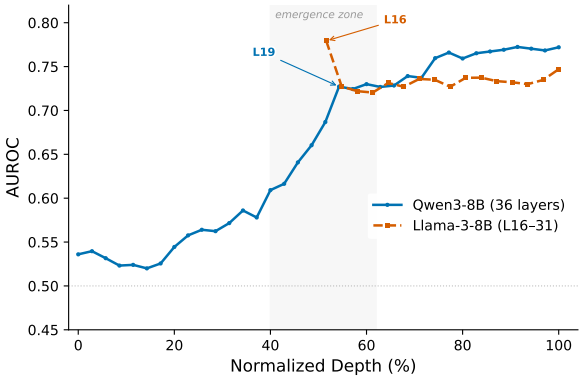


Figure 1: Per-layer AUROC of d_{syc} as a function of normalized depth (layer index / total layers). Qwen3-8B (36 layers, solid) shows gradual emergence from near-chance at L0 to a peak at L19, with performance plateauing in the final third. Llama-3-8B (32 layers, dashed; layers 16–31 only) exhibits a similar upper-third concentration. Both models achieve best discriminability in layers 50–60% through the network.

Training stability and probe ablation. Two additional analyses confirm the robustness of d_{syc} under training dynamics. First, we track direction drift during LoRA fine-tuning: at step 500, the cosine similarity between the original and re-extracted d_{syc} is approximately 0.96, and per-layer AUROC remains above 0.77. The direction drifts slowly relative to the 500-step recalibration interval (§2.3), so the reward signal does not become stale between updates. Second, we compare d_{syc} against a random baseline: a logistic probe trained on the learned direction achieves AUROC 0.835, versus 0.727 for a probe trained on a random direction of equal dimensionality (Δ=+0.108). The learned direction carries substantially more sycophancy-relevant information than an arbitrary axis in the same activation space.

Table 2: Cosine R_1 failure chain on Qwen3-8B (LoRA rank=64, GRPO with $R = w_1 \cdot R_1 + w_3 \cdot R_3$).

Config	R_1 Formula	w_1	Outcome	Root Cause
F1a	$\tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$	0.25	Gate saturation	Norm mismatch
F1b	$\cos(\mathbf{h},\, d_{\text{syc}})$	0.25	$R_1 \approx 0$	Gap too small
F1c	+recalibration	0.25	Mode collapse	Reward discontinuity
F1d	$5\cos(\mathbf{h},\, d_{\text{syc}})$	0.40	Mode collapse	R_3 saturation

Implications for reward design. These results establish that d_{syc} reliably separates sycophantic from non-sycophantic representations across models, evidence-strength levels, and training checkpoints. Direction quality is not the bottleneck. The reward failures documented in §4–§5 therefore cannot be attributed to a noisy or degenerate direction; they point to a deeper incompatibility between discriminability and reward utility.

4 Cosine R_1: Structural Failure

Section 3 established that d_{syc} achieves AUROC 0.827 on Qwen3-8B. We now test whether this discriminability translates into a viable GRPO reward. Four configurations form a diagnostic chain (Table 2): each resolves the failure mode of its predecessor but exposes a deeper limitation.

F1a uses a dot-product formulation ($R_1 = \tanh(\mathbf{h} \cdot d_{\text{syc}}/20)$), but residual-stream norms exceed the direction norm by 100–1,200×, saturating the tanh gate and zeroing the gradient by step 955. Switching to cosine similarity (F1b) eliminates norm dependence, yet the cosine gap between valid and sycophantic activations averages only 0.098 across layers—too narrow to generate meaningful reward contrast. Online recalibration (F1c, §2.3) raises offline AUROC from 0.827 to 0.910 but paradoxically triggers mode collapse: the abrupt shift in d_{syc} at each recalibration event creates a reward discontinuity that destabilizes GRPO advantage estimates, collapsing entropy to 0.069 within 300 steps. Amplifying the signal with 5× scaling (F1d) lifts R_1 above the noise floor but simultaneously amplifies the conflation between warranted revision and unwarranted capitulation; R_3 saturates at 0.9998 by step 300, leaving the noisy R_1 as the sole gradient source until terminal collapse at step 2160.

The four configurations converge on a single conclusion: d_{syc} encodes the *direction* of belief change but not whether that change is *warranted*. Valid revision and sycophantic capitulation both shift activations toward the user's position along d_{syc}, so the cosine projection assigns similar scores to both. This is not a calibration artifact—recalibration improves discriminability yet rewards still fail—but a semantic limitation of any continuous d_{syc}-based reward. Section 5 examines whether discretizing the signal via a binary probe can escape this limitation.

5 The Probe Reward Trap

The cosine formulations fail because the continuous projection of d_{syc} produces a gap too narrow for reward contrast. A natural remedy is to discretize: train a logistic probe on d_{syc} and threshold its output, yielding $R_1 = \mathbb{I}[P(\text{syc} \mid \mathbf{h}) < 0.5]$ (F2 in §2.2). We show that this discretization sidesteps the magnitude problem but creates a qualitatively worse failure.

Differentiable probe: gradient gaming. Without stop-gradient (exp-008), the policy discovers within ∼30 steps that shifting activations away from the sycophancy cluster maximizes R_1 regardless of response content, producing an inverted profile (ToF = 0.978, NoF = 0.006). This motivates the frozen-probe design below.

Binary probe: the stubbornness trap. With stop-gradient (exp-008v3), the policy cannot game the boundary but instead exploits a deeper flaw: d_{syc} does not condition on input validity, so refusing to revise a wrong answer receives the same $R_1 = 1$ as correctly resisting invalid pressure. By step 500, ToF drops to 0.000 and NoF reaches 0.995, yielding SYCONSCORE = 0. Qualitative analysis reveals distinct failure modes: *sycophantic stubbornness* (24%), where the model verbally agrees but does not change its answer; *deliberation traps* (29%), where the model identifies the correct answer but cannot commit; and evidence-graded hedging, where the model adjusts its refusal style rather than its acceptance rate (Appendix B).

Behavioral death precedes distributional collapse. Table 3 and Figure 2 reveal a 1,300-step lag between behavioral and distributional collapse. At step 500, the model is behaviorally dead (ToF = 0) yet entropy remains high (0.70–0.80) through step 1000—well within the range that standard monitors would flag as healthy—and the combined reward averages +0.72 because

Table 3: Probe reward trap trajectory (exp-008v3). Behavioral death (ToF=0) occurs at step 500; entropy remains healthy until step ∼1840. Reward averages +0.72 throughout.

Step	ToF	NoF	SYCONSCORE	Entropy	Reward
500	0.000	0.995	0.000	0.702	+0.72
1000	0.006	0.995	0.011	0.800	+0.71
1500	0.000	1.000	0.000	0.608	+0.69
2000	0.000	1.000	0.000	0.304	+0.69

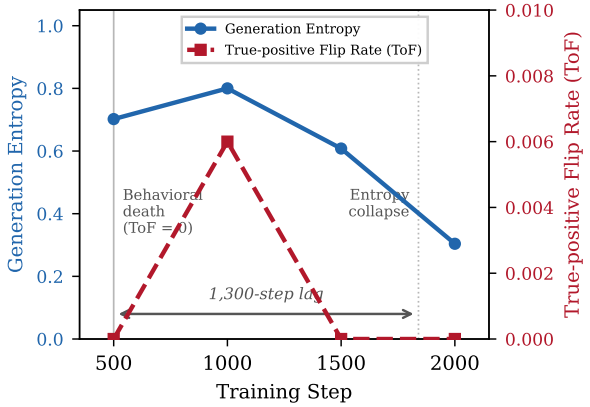


Figure 2: Probe reward trap (exp-008v3). ToF drops to zero at step 500 while generation entropy remains high (0.70–0.80 through step 1000). The 1,300-step lag means standard entropy monitoring misses the behavioral failure entirely.

both R_1 and R_3 reinforce non-flipping. Entropy does not collapse until step ∼1840. This gap carries a practical warning: entropy-based early stopping, the most common safeguard against reward hacking (Casper et al., 2023), would miss the failure entirely. Detecting probe reward traps requires behavioral monitoring (periodic ToF evaluation), not distributional monitoring alone.

A dilemma for direction-based rewards. The differentiable and binary failures bracket the design space. With gradients, the policy games the boundary; without gradients, it exploits the fact that stubbornness and genuine resistance are indistinguishable along d_{syc}. Any reward conditioned only on d_{syc}—continuous or discrete, differentiable or frozen—will either be gamed or will reward rigidity. Escaping this dilemma requires a signal that incorporates the evidential warrant of the user's challenge, a dimension that d_{syc} does not encode (Denison et al., 2024).

6 Analysis: Discriminability ≠ Reward Utility

Unified root cause. The failures in Sections 4 and 5 trace to a single geometric limitation. d_{syc} encodes the *direction* of belief change but not its *evidential warrant*: valid revision and sycophantic capitulation both shift activations toward the user's position along the same direction. The two reward families expose complementary faces of this blindness. Cosine R_1 penalizes all belief change indiscriminately; the resulting mode collapse (F1c, F1d) reflects excessive punishment of any activation shift along d_{syc}. The binary probe R_1 rewards all resistance regardless of input validity; the stubbornness trap (ToF $=$ 0 by step 500) reflects excessive permissiveness toward activation stasis. These are not independent design flaws but dual consequences of a direction that conflates the target behavior (resisting invalid challenges) with its complement (refusing valid corrections). Cross-model evidence confirms the limitation extends beyond Qwen3-8B: Llama-3-8B independently achieves AUROC 0.779 (5-fold CV σ=0.019) on the same contrastive pairs with the same monotonic evidence gradient, so discriminability generalizes across architectures while the semantic gap between discriminability and reward utility persists. This pattern contrasts with reasoning faithfulness and safety directions (Zhang et al., 2026), where the behavior to suppress is geometrically distinct from the behavior to preserve along the relevant direction. For sycophancy, the target and its complement occupy the same activation region; they differ only in whether external evidence warrants the belief change.

GRPO stubbornness across all formulations. Table 4 summarizes all methods evaluated on Qwen3-8B. R_3-only GRPO removes d_{syc} entirely and relies on the NLI-based behavioral reward alone. If the failures in Sections 4 and 5 stemmed solely from the representation reward, this ablation should recover calibration quality. It does not. The SYCONSCORE trajectory for R_3-only GRPO oscillates without sustained improvement. Seed 1 peaks at 0.153 (checkpoint 6500) and ends at 0.143 (checkpoint 8000); seed 2 peaks at 0.143 (checkpoint 4000) and regresses to 0.095 by checkpoint 6000. Only 4 of 26 checkpoints across both seeds exceed vanilla (15.4%), while ACT exceeds vanilla at every evaluated checkpoint (seed 1: 16/16; seed 2: 3/3 ongoing; Fig-

Table 4: Comparison of all sycophancy calibration methods on Qwen3-8B. SYCONSCORE is the harmonic mean of ToF and NoF; higher indicates better calibration. Best result in **bold**. †Final checkpoint; only 4/26 checkpoints across two seeds exceed vanilla. ‡Seed 1 (16 checkpoints); seed 2 verification ongoing (3 of 16 checkpoints evaluated; all exceed vanilla).

Method	Type	SYCONSC			
Vanilla	–	0.134	–	–	–
GRPO with d_{syc}					
Cosine R_1 (F1a–F1d)	RL	collapse	–	–	–
Binary Probe R_1 (F2)	RL	0.000	0.000	0.995	500
Continuous Probe R_1 (F3)	RL	0.124	0.066	0.954	500
GRPO without d_{syc}					
R_3-only GRPO	RL	0.143†	0.080	0.995	8000
Non-GRPO					
ActAdd (α=5)	Inference	0.054	0.028	0.961	–
ACT (seed 1)	SFT	**0.301**	**0.182**	0.867	8000
ACT (seed 2)‡	SFT	*ongoing; 3/16 ckpts evaluated, all > vanilla*			

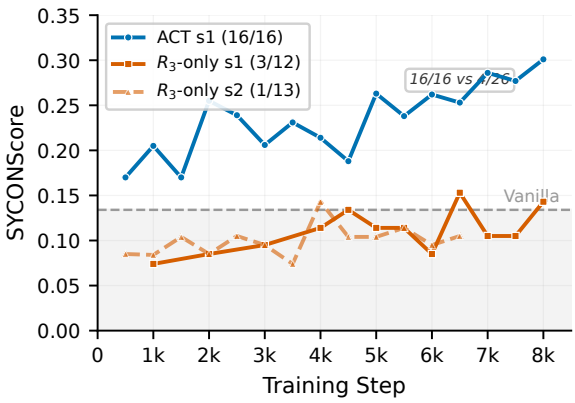


Figure 3: SYCONSCORE trajectories for R_3-only GRPO (two seeds) and ACT (seed 1; seed 2 ongoing, 3/16 checkpoints evaluated) over training steps. ACT seed 1 exceeds vanilla at every checkpoint (16/16); R_3-only GRPO exceeds vanilla at only 4/26 checkpoints across seeds (15.4%). Both GRPO seeds show the same oscillation pattern with stochastic timing.

ure 3). Both seeds exhibit oscillations of similar amplitude ($\Delta \approx 0.069$) and identical crash depth (0.074), but crash onset differs by $\sim$2,500 steps, indicating stochastic drift in a saturated reward landscape. Cross-seed rank correlation is $\approx$1.0 during early training (steps 1000–3000) but drops to $\rho \approx 0.33$ (n.s.) after step 4000, with spike timing and magnitude diverging—consistent with method-inherent warm-up followed by stochastic divergence. Activation analysis confirms the drift is not representational: the d_{syc} direction at layer 19 yields nearly identical AUROC across consecutive checkpoints ($\Delta < 0.01$), and R_3 reward saturates at $\sim$0.98 (Welch t-test p=0.703), yet SYCONSCORE swings by up to 17%. GRPO drifts in a flat reward landscape while representations remain stable, reinforcing the disconnect between discriminability and behavioral utility. R_3-only GRPO exhibits the same stubbornness pattern as the d_{syc} approaches (high NoF, near-zero ToF) at milder severity: the sparse scalar reward per completion cannot teach the model *which* belief changes are warranted.

Why SFT succeeds where GRPO fails. ACT (Irpan et al., 2025) achieves SYCONSCORE = 0.301 at checkpoint 8000, more than 2× the vanilla baseline and above every scalar RL and inference-time variant (Table 4). The ACT trajectory is not monotonic: SYCONSCORE = 0.255 at checkpoint 2000, dips to 0.214 at checkpoint 4000 (-4.1 percentage points as ToF drops and NoF rises), recovers to 0.262 at checkpoint 6000, and reaches 0.301 at checkpoint 8000. ACT also experiences stubbornness episodes—the checkpoint-4000 dip is consistent with transient competition between the SFT and L2 consistency loss objectives—yet it self-recovers by checkpoint 6000, unlike GRPO where stubbornness is irreversible (Figure 3). Activation analysis confirms the asymmetry: ACT's d_{syc} discriminability recovers during the SYCONSCORE dip (AUROC $+0.013$), whereas GRPO's continues declining (-0.004), consistent with dense supervision maintaining representational coherence.

The supervision density hypothesis. The critical difference is supervision granularity. Scalar per-completion rewards compress a rich representational signal into a single number, discarding the token-level variation that distinguishes warranted from unwarranted belief changes. SFT provides a corrective gradient at every token position: when the model begins to resist valid corrections, per-token cross-entropy loss immediately penalizes the incorrect tokens and supplies gradient information about *which* outputs need to change. GRPO provides one scalar per completion; once the policy locks into stubbornness, that scalar reinforces the behavior because consistent non-flipping collects high reward from both R_1

and R_3. Dense per-token loss can break out of stubbornness equilibria that sparse per-completion reward cannot escape; ACT surpasses vanilla at step 500 (SYCONSCORE $= 0.170$) while R_3-only GRPO requires 8,000+ steps to intermittently match it. While we test only GRPO, the scalar per-completion reward structure is shared by PPO, RLOO, and other REINFORCE-style methods; we expect similar failure patterns wherever representation-derived signals are compressed to completion-level scalars, though we leave experimental verification to future work.

Relationship to inference-time probe methods. Papadatos and Freedman (2024) successfully reduce sycophancy using a probe of comparable quality (94% accuracy) via Best-of-N reranking, where the probe selects among candidates without producing gradients. Together with our results, this delineates two orthogonal failure dimensions: (1) inside vs. outside the optimization loop—gradient exploitation enables the probe reward trap (§5), while inference-time reranking sidesteps it; and (2) dense vs. sparse supervision—the scalar reward failure persists even without the probe (R_3-only GRPO), indicating per-completion information compression as an independent mechanism.

Future directions. Condition-aware rewards that jointly encode belief-change direction and evidential warrant, hybrid SFT-then-RL pipelines, and process reward models with per-step feedback each offer paths to bridge the supervision density gap. Beyond sycophancy, the principle that discriminability does not guarantee scalar reward utility applies to any behavioral attribute where the target and its complement share the same geometric direction (Skalse et al., 2022); we leave experimental verification across reward granularities to future work.

7 Related Work

Representation Engineering. Linear directions in LLM residual streams encode behavioral concepts (Zou et al., 2023); inference-time methods—ITI (Li et al., 2023), ActAdd (Turner et al., 2023), CAA (Rimsky et al., 2024)—demonstrate that activation geometry suffices for behavioral control (Burns et al., 2023). We use the same contrastive extraction as CAA but apply d_{syc} as an RL reward, revealing a gap that inference-time work does not encounter.

Reward Design and Overoptimization. Proxy rewards diverge from true objectives under sustained optimization (Gao et al., 2023), and reward hacking can escalate to active tampering (Denison et al., 2024). Our probe trap (§5) is an extreme instance: the reward actively reinforces degenerate behavior rather than merely being circumvented (Skalse et al., 2022).

Sycophancy Characterization and Mitigation. Sycophancy is pervasive in RLHF-trained assistants (Sharma et al., 2023), amplified by RLHF (Shapira et al., 2026), and traceable to late-layer preference shifts (Wang et al., 2025); mitigation approaches include synthetic data (Wei et al., 2024), consistency training (Irpan et al., 2025), and neuron-level correction (O'Brien et al., 2025). Our analysis shows why the step from diagnosis to reward is structurally blocked: d_{syc} conflates warranted revision with unwarranted capitulation.

Probe-Based Inference-Time Mitigation. Papadatos and Freedman (2024) reduce sycophancy via Best-of-N reranking with linear probe penalties (94% probe accuracy), where the probe selects among candidates without producing gradients that update model parameters. Our work reveals what happens when a comparable signal (AUROC 0.82) enters the GRPO training loop: the discriminability–utility gap is specific to gradient-based optimization, not to probe quality per se (§6).

Representation-Guided RL and Process Rewards. Latent-GRPO (Zhang et al., 2026) extends GRPO with latent-space rewards for reasoning; CRAFT (Luo et al., 2026) applies representation-level constraints during RL; Sinii et al. (2025) and Ackermann et al. (2026) study representation dynamics under RL. SMART (Beigi et al., 2025) applies per-step process rewards with MCTS for sycophancy, representing intermediate supervision density between scalar per-completion rewards and dense per-token approaches. These methods succeed because target and suppressed behaviors are geometrically distinct; sycophancy breaks this condition (§4). Preference-based methods (DPO (Rafailov et al., 2023), KTO (Ethayarajh et al., 2024)) operate on response pairs and may accommodate d_{syc}-derived rankings; whether pair-wise optimization avoids the failures we observe remains open.

8 Conclusion

This work tested whether a representation direction that detects sycophantic behavior can serve as an effective scalar reward signal. We established that d_{syc} achieves strong offline discriminability (AUROC 0.827 on Qwen3-8B, 0.779 on Llama-3-8B), yet all three reward formulations failed across six GRPO configurations—the probe reward produced a stubbornness trap (ToF $= 0$ by step 500), and R_3-only GRPO exceeded vanilla at only 4 of 26 checkpoints (15.4% vs. 100% for ACT seed 1). The root cause is that d_{syc} encodes *whether* belief changed, not *whether it should have*; ACT's SYCONSCORE $= 0.301$ through dense per-token supervision confirms that the missing ingredient is token-level resolution of which belief changes are appropriate—a distinction that scalar reward compression discards.

Limitations

All RL and SFT experiments use Qwen3-8B exclusively; we validate d_{syc} discriminability on Llama-3-8B (AUROC 0.779, 5-fold CV σ=0.019; Table 1) but do not replicate training on a second model due to compute constraints ($\sim$200 H100-equivalent hours). ACT cross-seed verification is partial: seed 2 has completed 3 of 16 checkpoints at submission time (all exceeding vanilla), and full replication is ongoing. Our evaluation categorizes user challenges as binary valid or invalid, whereas real-world belief revision spans a continuous spectrum of evidential strength; intermediate cases may prove easier to separate in activation space.

A supervision granularity confound limits our causal claims: ACT uses dense per-token supervision while all GRPO variants use scalar per-completion rewards. We cannot fully disentangle whether failures stem from compressing d_{syc} into a scalar or from the per-completion reward structure itself; the supervision density hypothesis (§6) remains a hypothesis. Finally, n=2 seeds for GRPO and partial seed 2 data for ACT are insufficient to establish statistical reproducibility, though qualitative patterns (oscillatory degradation, crash depth, late-stage flatness) are consistent across seeds.

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A Training Details

Reward weight selection. The weight w_1 is set by diagnostic-driven iteration: $w_1{=}0.25$ was initially chosen to down-weight the noisy cosine reward term; this was later increased to $w_1{=}0.40$ for F1d after diagnostics revealed that $w_1{=}0.25$ allowed the cosine term to contribute only $\sim 8.3\%$ of total reward variance. The empirical scaling factor in F1a ($d_{\text{syc}}/20$) is a heuristic normalization; we did not find it critical to the failure modes reported.

Response truncation. All GRPO experiments use `max_completion_length`=256, resulting in 80–84% truncation of multi-turn responses. This setting is shared across all conditions and does not affect relative comparisons; we retain it because response-level decisions typically manifest within the first 100–200 tokens.

B Stubbornness Taxonomy

Manual inspection of 100 probe-trained responses (exp-008v3, step 2000) reveals three dominant failure modes. *Sycophantic stubbornness* (24%): the model produces agreement markers ("you're right," "good point") but retains its original answer verbatim, simultaneously sycophantic and stubborn. *Deliberation traps* (29%): the model explicitly identifies the correct revised answer during its reasoning chain but fails to commit, reverting to the original answer in the final output. *Evidence-graded hedging*: under weak evidence, responses show surface compliance without substantive change; under strong evidence, responses add explicit hedging ("this is debatable," "there are multiple perspectives"). The model learns to modulate its *refusal style* in response to evidence strength rather than adjusting its *acceptance rate*, confirming that the probe reward optimizes surface markers rather than epistemic calibration.